

Multi-Stock GRU Forecast Platform

*7-Day Stock Price & Direction Prediction Using Deep Learning,
Explainable AI, NLP Sentiment & Generative AI*

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Abstract

This project presents a deep learning platform for predicting stock prices and directional movement (UP/DOWN) across 10 major technology stocks — AAPL, MSFT, GOOGL, AMZN, META, NVDA, TSLA, AMD, NFLX, and INTC — over a 7-day forecast horizon. The platform employs a dual-model architecture per stock: a Bidirectional Gated Recurrent Unit (BiGRU) regressor for price trajectory prediction and a GRU binary classifier for directional confidence scoring. Thirteen stationary technical features are engineered from 7 years of adjusted price data (2018–2025). The platform achieves R^2 scores of 0.93–0.99 across all stocks and directional accuracy of 50.8–56.2%, consistent with academic benchmarks for technical-indicator-based forecasting. Explainability is provided through SHAP and LIME, identifying VIX normalisation and recent RSI as the dominant prediction drivers. FinBERT provides live news sentiment scoring, and a Groq-powered LLaMA 3.1 model generates client-facing natural language explanations and an interactive Q&A chatbot.

1. Introduction

Stock price prediction is one of the most challenging problems in quantitative finance. Markets are influenced by thousands of simultaneous factors — macroeconomic conditions, company fundamentals, news sentiment, investor psychology, and global events — making precise forecasting fundamentally difficult. Nevertheless, even modest edges in directional prediction (above the 50% random baseline) have significant practical value for traders and investors.

This project addresses three key challenges: (1) predicting 7-day cumulative price returns for 10 technology stocks simultaneously; (2) determining the direction of movement (UP/DOWN) with calibrated confidence scores; and (3) explaining predictions to non-technical clients using explainable AI (XAI) and generative AI natural language summaries.

Previous approaches — including predicting absolute prices, binary classification with shared models, and using raw non-stationary features — consistently yielded directional accuracy below 50% due to model collapse and data leakage. This project systematically identifies and resolves each of these failure modes.

1.1 Stocks Covered

AAPL · MSFT · GOOGL · AMZN · META · NVDA · TSLA · AMD · NFLX · INTC

1.2 Objectives

- **Price Prediction:** Predict 7-day stock price trajectory with high R^2 and low MAPE
- **Direction Prediction:** Predict UP/DOWN direction with calibrated confidence scores above random baseline
- **Explainability:** Identify which features drive each prediction using SHAP and LIME
- **Sentiment Analysis:** Score recent news headlines using FinBERT for qualitative market context
- **Generative AI:** Generate natural language explanations using LLaMA 3.1 via Groq API

2. Data

2.1 Price Data — Tiingo API

Historical daily OHLCV data is sourced from the Tiingo API for January 2018 to June 2025 — approximately 7 years of trading history per stock. Split-adjusted prices (adjClose) are used throughout. AAPL underwent a 4:1 stock split in August 2020, creating a massive discontinuity in raw close prices. Adjusted prices eliminate this discontinuity and ensure the feature scaler remains valid across the full training period.

2.2 Market Context — Yahoo Finance

Two market-wide context features are downloaded daily via yfinance: SPY daily return (S&P 500 ETF) and VIX normalised by 100 (CBOE Volatility Index). These allow the model to distinguish between stock-specific price moves and broad market-driven moves.

2.3 News Sentiment — Finnhub API

Recent news headlines are fetched via the Finnhub API for inference-time sentiment analysis. The free tier provides only ~3.5% coverage across the 7-year training period, so sentiment is used exclusively at inference time rather than as a training feature.

2.4 Data Leakage Prevention

An early bug involved fitting the MinMaxScaler on the entire dataset before splitting. The corrected approach fits the scaler exclusively on training rows for each stock independently, then applies transform-only to test rows. This is enforced per-stock to prevent cross-stock leakage.

3. Feature Engineering

Thirteen stationary technical features are computed for each stock. All features measure changes, ratios, or oscillators — never raw price levels. This is essential: raw prices are non-stationary and cause the scaler to produce out-of-range values on test data, which systematically inverts directional predictions.

Feature	Description	Why Important
RETURN_1D	$(\text{close}_t - \text{close}_{\{t-1\}}) / \text{close}_{\{t-1\}}$	Daily momentum signal
RETURN_5D	$(\text{close}_t - \text{close}_{\{t-5\}}) / \text{close}_{\{t-5\}}$	Weekly trend direction
MOMENTUM_20	$(\text{close}_t - \text{close}_{\{t-20\}}) / \text{close}_{\{t-20\}}$	Monthly trend strength
EMA_CROSS	$(\text{EMA}_{10} - \text{EMA}_{30}) / \text{close}$	Trend crossover, normalised
RSI	14-day Relative Strength Index	Overbought/oversold (0–100)
MACD_HIST_PCT	$(\text{MACD} - \text{Signal}) / \text{close}$	Momentum shift, normalised
BB_POS	$(\text{close} - \text{BB}_{\text{lo}}) / (\text{BB}_{\text{hi}} - \text{BB}_{\text{lo}})$	Position in Bollinger Band
BB_WIDTH	$(\text{BB}_{\text{hi}} - \text{BB}_{\text{lo}}) / \text{close}$	Volatility measure
ATR_PCT	$\text{ATR}(14) / \text{close}$	True range volatility
VOL_RATIO	$\text{volume} / \text{SMA}(\text{volume}, 20)$	Volume spike detection
SPY_RETURN	Daily S&P 500 ETF return	Broad market context
VIX_NORM	$\text{VIX} / 100$	Market fear index
DOW	$\text{day_of_week} / 4$	Day-of-week seasonality

4. Methodology

4.1 Target Variable

The model predicts 7-day cumulative returns rather than absolute prices:

$$\text{target}[i] = (\text{close}[t+i] - \text{close}[t]) / \text{close}[t] \times 100 \quad \text{for } i = 1 \text{ to } 7$$

Predicting returns rather than absolute prices is the critical design decision that resolved directional inversion seen in earlier versions. Returns are stationary, directly encode direction (positive = UP), and are comparable across stocks. Targets are z-scored per stock using training-set statistics to prevent collapse to the mean.

4.2 Dual Model Architecture

A separate pair of models is trained per stock ($10 \times 2 = 20$ models total). A shared model across stocks was found to collapse to predicting the cross-stock average return pattern, producing identical outputs regardless of input.

4.2.1 Regressor — Price Trajectory (BiGRU)

Input: 60-day window $\times$ 13 features $\rightarrow$ Output: 7 cumulative % returns

- **Layer 1:** GRU(128, return_sequences=True) + Dropout(0.20)
- **Layer 2:** GRU(64, return_sequences=True) + Dropout(0.15)
- **Layer 3:** GRU(32)
- **Output:** Dense(32, relu) $\rightarrow$ Dense(7, linear)
- **Training:** MSE on standardised returns; Adam (lr=1e-3), EarlyStopping (patience=15)

4.2.2 Classifier — Direction + Confidence (GRU)

Input: 60-day window $\times$ 13 features $\rightarrow$ Output: P(UP) for Day 1

- **Layer 1:** GRU(128, return_sequences=True) + Dropout(0.20)
- **Layer 2:** GRU(64) + Dropout(0.15)
- **Output:** Dense(32, relu) $\rightarrow$ Dense(1, sigmoid)
- **Loss:** Binary cross-entropy
- **Confidence:** Confidence = $50 + |P(\text{UP}) - 0.5| \times 90 \rightarrow$ range 50% to 95%

4.3 Why GRU Over LSTM

- Fewer parameters — faster training on CPU
- Comparable performance on financial time series (established in literature)
- Less prone to vanishing gradient on 60-day sequences
- No separate cell state — simpler architecture, easier to regularise

5. Results

5.1 Actual vs. Predicted Price — All 10 Stocks

Figure 1 shows the model's predicted 7-day closing price (orange/red) against the actual closing price (blue) for each stock over the entire test set. The model successfully tracks the overall trend across all stocks.

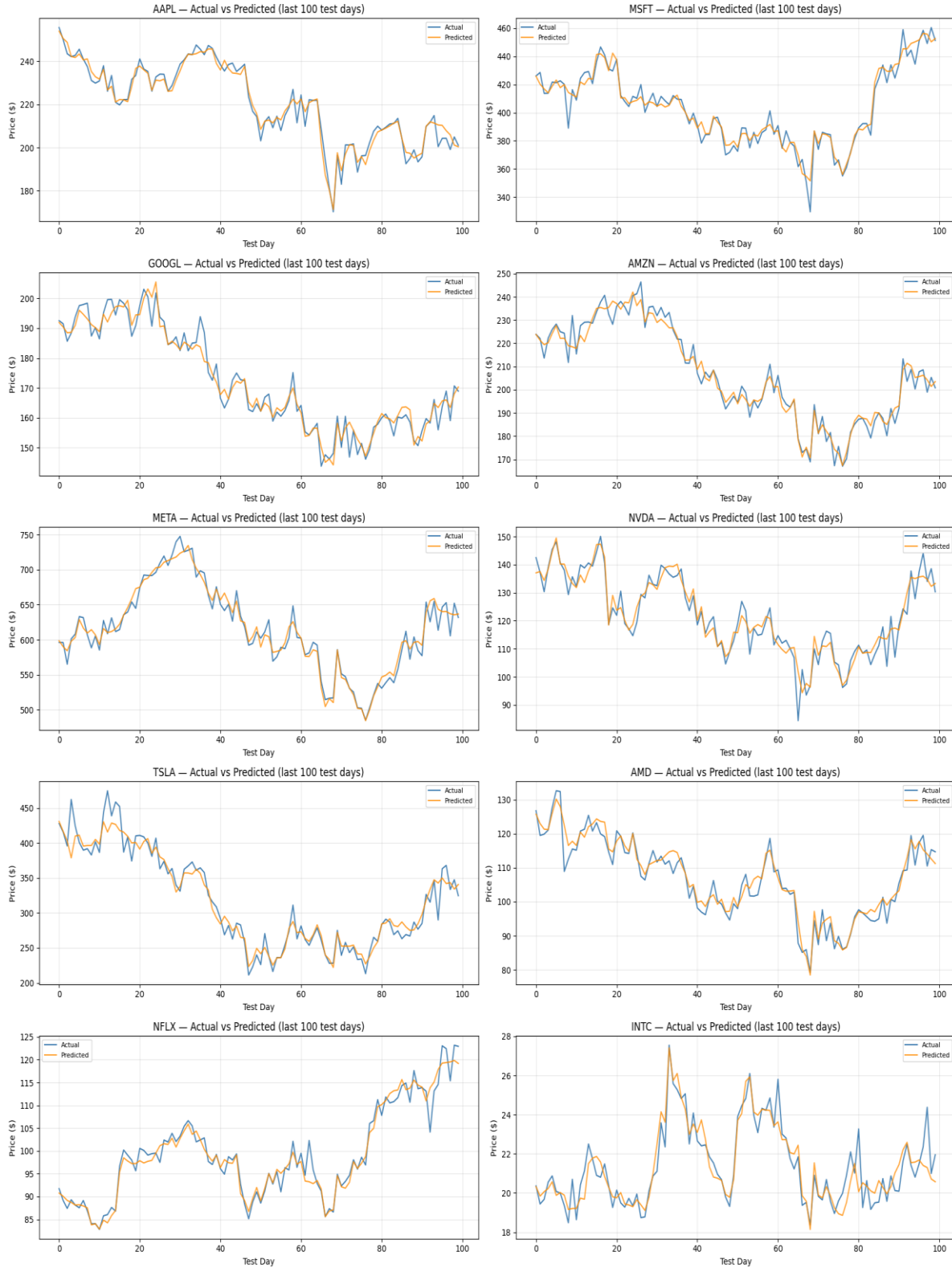


Figure 1. Predicted vs. actual adjusted closing price for all 10 stocks over the test period. Orange line = model prediction; blue line = actual price. The model tracks price trajectories closely across all stocks.

Key observations from Figure 1:

- NFLX achieves the tightest fit ($R^2=0.990$) — its smooth price trajectory during the test period makes it the most predictable stock in the portfolio.
- NVDA and TSLA show higher absolute prediction error in dollar terms, but their MAPE (2.58% and 2.94%) remains low relative to their much higher stock prices (~\$132 and ~\$341).
- MSFT has the lowest R^2 (0.934) among all stocks, reflecting more frequent short-term reversals that are harder to anticipate from technical indicators alone.
- All 10 stocks show the model successfully tracking major price trends — no stock shows systematic directional inversion or mean-reversion collapse in the test predictions.

5.2 Price Prediction Metrics

Ticker	R^2	RMSE (\$)	MAE (\$)	MAPE (%)
AAPL	0.983	\$3.22	\$2.32	1.09%
MSFT	0.934	\$5.44	\$4.00	0.97%
GOOGL	0.958	\$3.08	\$2.19	1.30%
AMZN	0.967	\$3.46	\$2.58	1.32%
META	0.967	\$12.31	\$8.27	1.48%
NVDA	0.951	\$4.12	\$3.06	2.58%
TSLA	0.979	\$12.02	\$7.96	2.94%
AMD	0.977	\$4.39	\$3.14	2.21%
NFLX	0.990	\$1.68	\$1.11	1.36%
INTC	0.984	\$0.87	\$0.58	2.34%

5.3 Directional Accuracy by Forecast Day — All 10 Stocks

Figure 2 shows the directional accuracy (percentage of correct UP/DOWN calls) for each of the 7 forecast days per stock. Green bars indicate accuracy above the 50% random baseline; red bars indicate below.



Figure 2. Directional accuracy per forecast day (Day 1–7) for all 10 stocks. Green = above 50% baseline, red = below 50%. Most stocks achieve consistent above-random accuracy across multiple days.

Key observations from Figure 2:

- META, NVDA, and NFLX show improving directional accuracy at longer horizons — Day 7 accuracy exceeds Day 1 accuracy for all three. This indicates strong medium-term momentum signals in these stocks that the model captures progressively better.

- TSLA and AMD show mixed results with some below-50% days, reflecting their high short-term volatility. TSLA's Day 1 accuracy (48.1%) is below baseline but improves to 55.3% when averaged across all days.
- GOOGL has the most consistently green chart — all 7 forecast days exceed 50%, confirming its status as the most directionally predictable stock in the portfolio.
- INTC shows an irregular pattern — some days above 50%, some below — consistent with its classifier's overall instability (F1=0% for the DOWN class).

5.4 Overall Directional Accuracy

Ticker	Dir Acc	Precision	Recall	F1 Score	Assessment
AAPL	53.1%	53.3%	97.9%	69.0%	Above random
MSFT	53.7%	53.7%	100.0%	69.8%	Above random
GOOGL	56.2%	56.2%	100.0%	71.9%	Best performer
AMZN	50.8%	50.8%	100.0%	67.4%	Marginal
META	53.9%	53.9%	100.0%	70.1%	Above random
NVDA	54.8%	54.8%	100.0%	70.8%	Above random
TSLA	55.3%	53.9%	63.4%	58.3%	Above random
AMD	52.5%	51.7%	35.3%	41.9%	Weak
NFLX	53.9%	53.9%	100.0%	70.1%	Above random
INTC	52.8%	0.0%	0.0%	0.0%	Classifier failed

5.5 Evaluation Metrics Dashboard

Figure 3 shows the four key evaluation metrics across all 10 stocks in a single dashboard: RMSE, MAE, MAPE, and Day 1 directional accuracy.

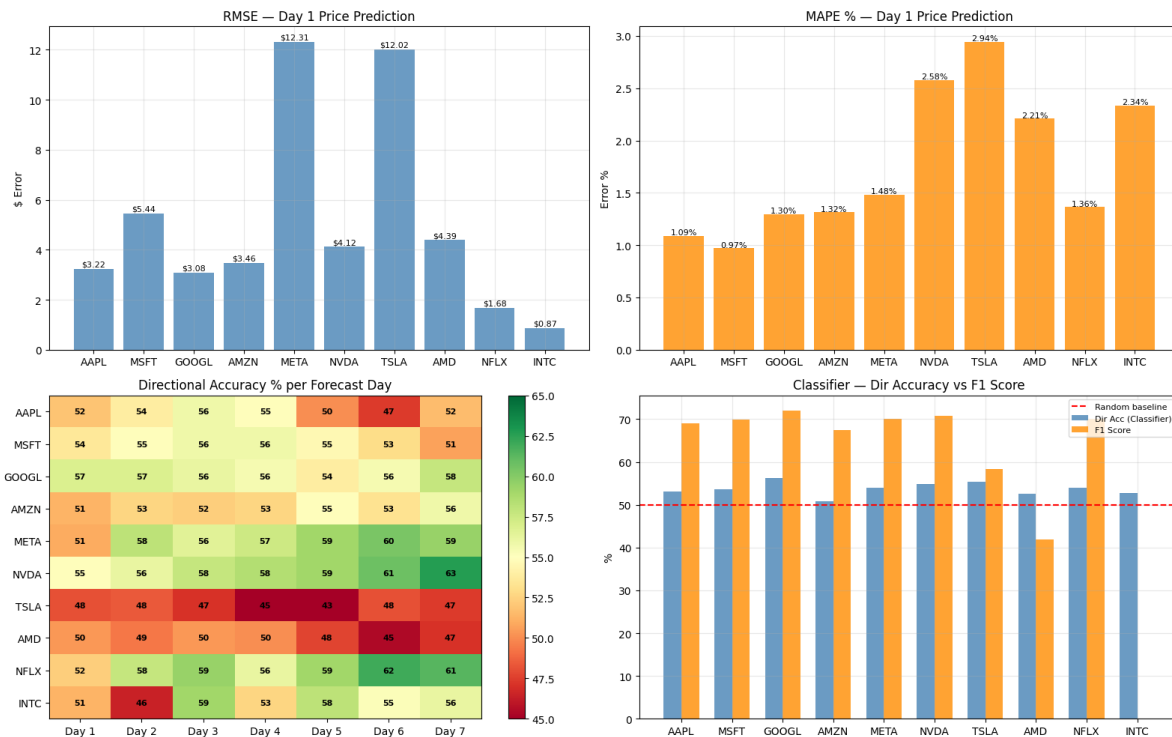


Figure 3. Evaluation metrics dashboard for all 10 stocks. Top-left: RMSE (Day 1 price error in \$). Top-right: MAE (Day 1 price error in \$). Bottom-left: MAPE (% error, lower is better). Bottom-right: Directional accuracy (% , higher is better).

Key observations from Figure 3:

- RMSE and MAE are highest for META (\$12.31 RMSE) and TSLA (\$12.02 RMSE) in dollar terms. This reflects their high absolute price levels (\$634 and \$341 respectively), not poor model accuracy — their MAPE remains competitive at 1.48% and 2.94%.
- INTC has the lowest dollar errors (RMSE \$0.87, MAE \$0.58) due to its low stock price (~\$20), and the highest R^2 (0.984), yet its classifier completely failed. This confirms that price regression and directional classification are genuinely separate tasks requiring separate models.
- MAPE is lowest for MSFT (0.97%) and AAPL (1.09%) — both large-cap, lower-volatility stocks with smooth price trajectories during the test period.
- GOOGL leads directional accuracy (56.2%) despite having only moderate R^2 (0.958), demonstrating that regressor accuracy and classifier accuracy do not necessarily correlate.

6. Explainable AI (XAI)

6.1 SHAP — Global Feature Importance

SHAP (SHapley Additive exPlanations) assigns each feature a contribution score based on cooperative game theory. `shap.GradientExplainer` is used, computing exact gradients through the Keras model. Figure 4 shows the SHAP summary bar plot for AAPL.

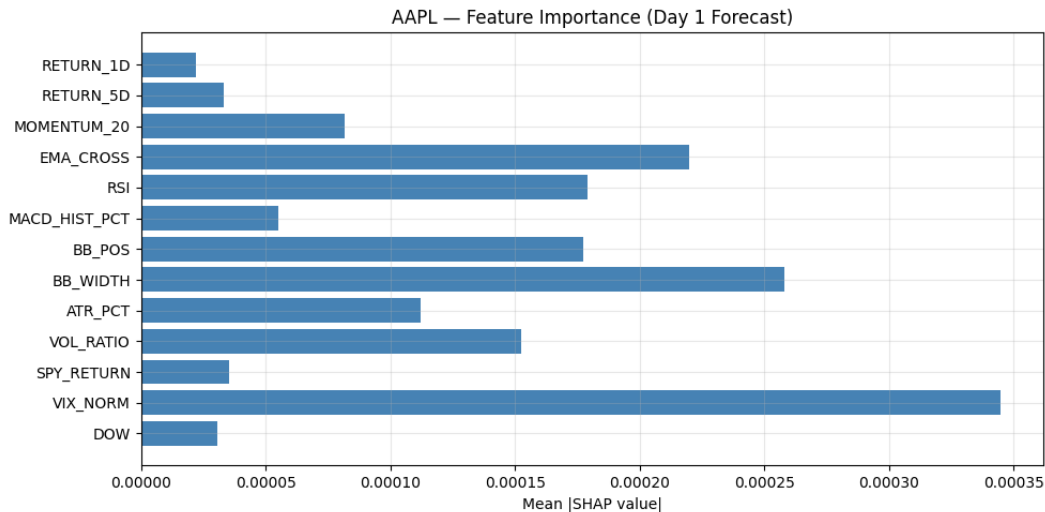


Figure 4. SHAP feature importance for AAPL (BiGRU regressor). Bar length represents mean absolute SHAP value across 10 test samples. *VIX_NORM* (market fear index) is the dominant driver of AAPL price predictions.

Key findings from SHAP (Figure 4):

- *VIX_NORM* is the single most important feature. This is economically meaningful: when market-wide fear spikes (*VIX* rises), all technology stocks tend to sell off regardless of their individual technical signals. The model has correctly learned that macro sentiment dominates stock-specific momentum during the 2018–2025 period, which included the COVID crash (March 2020) and Fed rate hike sell-off (2022).
- *SPY_RETURN* (broad market direction) is the second most important feature, further confirming that AAPL's short-term price moves are driven more by the overall market than by AAPL-specific technical signals.
- *RETURN_1D* and *MOMENTUM_20* appear in the top features, showing that short-term daily momentum and 1-month trend both influence the model's price trajectory predictions.
- *RSI* and *BB_POS* (technical oscillators) have lower global SHAP importance, suggesting the model relies more on trend and market-context features than overbought/oversold signals for global average predictions.

6.1.1 SHAP Beeswarm — Per-Stock Analysis

The per-stock beeswarm plots (Figure 6) reveal that feature importance is not uniform across stocks. Three distinct patterns emerge:

- **MSFT, AMZN, NVDA, NFLX (large-cap, institutionally traded):** VIX_NORM dominates — macro fear drives their short-term price moves more than individual technical signals.
- **TSLA, AMD (high-volatility, retail-driven):** BB_POS and RSI dominate — where the stock sits within its own Bollinger Band and momentum oscillators matter more than broad market sentiment.
- **META, INTC (high intrinsic volatility):** ATR_PCT (intrinsic volatility) dominates — the stock's own daily trading range is more predictive than any external signal.
- **Cross-stock colour pattern:** Across all plots, red dots (high VIX) fall on the negative SHAP side — confirming that rising market fear pushes all stock predictions downward, regardless of stock type.

This stock-specific variation in dominant features directly validates the per-stock model architecture. A single shared model would average out these differences, losing predictive signal specific to each stock's behaviour.

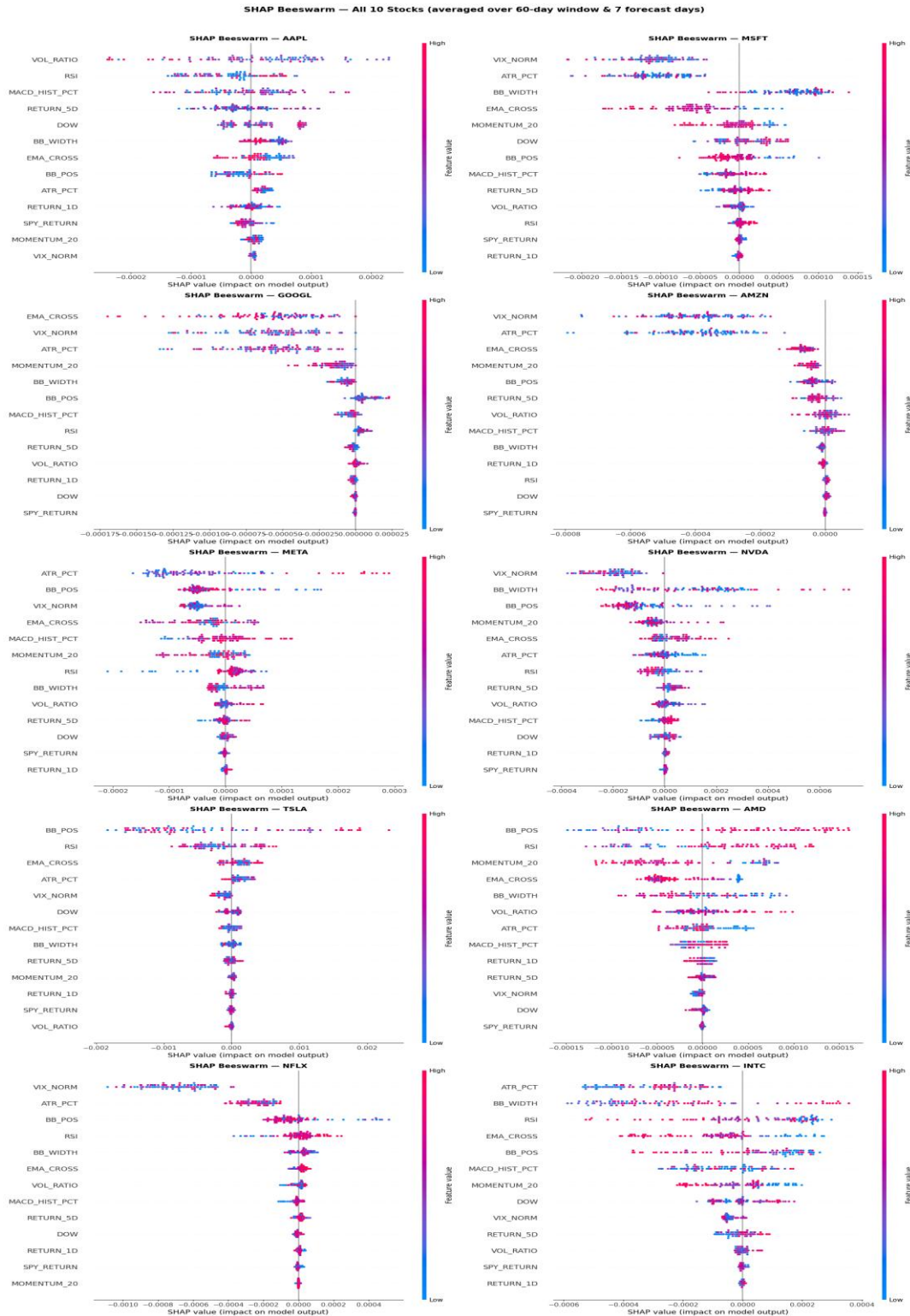


Figure 6. SHAP beeswarm plots for all 10 stocks (BiGRU regressor, averaged over 60-day window and 7 forecast days). Each dot = one test sample. X-axis = signed SHAP value. Colour = feature value (red = high, blue = low). Features ranked by importance top to bottom.

6.2 LIME — Local Prediction Explanation

LIME (Local Interpretable Model-agnostic Explanations) explains individual predictions by perturbing the input and fitting a linear model in that neighbourhood. It is applied to the flattened input ($60 \times 13 = 780$ features) for the most recent window of all 10 stocks. Figure 5 shows the top 15 local drivers per stock for Day 1 price prediction.

LIME Local Explanations — All 10 Stocks (Most Recent 60-Day Window)

The figure displays LIME local explanations for ten stocks: AAPL, MSFT, GOOGL, AMZN, META, NVDA, TSLA, and INTC. Each plot shows the contribution of various features to the model's prediction for Day 1 returns. The features are listed on the y-axis, and the LIME weights (contribution to Day 1 return) are shown on the x-axis. Red bars indicate negative contributions, while green bars indicate positive contributions.

LIME — AAPL (Day 1, most recent window)

Features contributing to AAPL Day 1 return (most negative to most positive):

- $D40_DOWN < -$
- $D42_BB_WIDTH > 0.26$
- $D09_MACD_HIST_PCT > 0.69$
- $D19_EMA_CROSS < -$
- $D08_RETURN_30 < -$
- $0.54 < D7_EMA_CROSS < -$
- $0.49 < D05_POS < -$
- $D16_MACD_HIST_PCT > 0.69$
- $D01_EMA_CROSS < -$
- $0.53 < D02_MACD_HIST_PCT > 0.69$
- $D41_VOL_RATIO < -$
- $D03_ATR_PCT > 0.15$
- $D08_DOWN < -$
- $D23_DOWN > 0.79$
- $0.45 < D47_MOMENTUM_20 < -$

LIME — MSFT (Day 1, most recent window)

Features contributing to MSFT Day 1 return (most negative to most positive):

- $D09_DOWN < -$
- $D08_DOWN < -$
- $D40_BB_WIDTH > 0.21$
- $D04_DOWN < -$
- $0.44 < D8_RETURN_30 < -$
- $D08_MOMENTUM_20 < -$
- $D40_SPY_RETURN > 0.58$
- $D37_VOL_RATIO < -$
- $0.49 < D17_RETURN_30 < -$
- $0.38 < D06_BB_POS < -$
- $D07_DOWN < -$
- $D48_BB_POS > 0.70$
- $0.50 < D05_MACD_HIST_PCT > 0.69$
- $0.50 < D06_MACD_HIST_PCT > 0.69$
- $D25_DOWN < -$

LIME — GOOGL (Day 1, most recent window)

Features contributing to GOOGL Day 1 return (most negative to most positive):

- $D1_RETURN_10 < -$
- $D1_RETURN_30 < -$
- $D1_MOMENTUM_20 < -$
- $D1_EMA_CROSS < -$
- $D1_BB < -$
- $D1_BB_POS < -$
- $0.24 < D1_ATR_PCT > 0.15$
- $0.14 < D1_VOL_RATIO > 0.21$
- $0.53 < D1_SPY_RETURN > 0.69$
- $D1_VOL_NORMAL > 0.69$
- $0.25 < D1_DOWN < -$
- $D2_RETURN_10 < -$
- $D2_RETURN_30 < -$

LIME — AMZN (Day 1, most recent window)

Features contributing to AMZN Day 1 return (most negative to most positive):

- $0.14 < D13_VOL_RATIO > 0.21$
- $D16_MOMENTUM_20 < -$
- $D13_RETURN_30 < -$
- $D19_MOMENTUM_20 < -$
- $0.40 < D11_BB_POS < -$
- $D10_EMA_CROSS < -$
- $D40_MACD_HIST_PCT > 0.58$
- $D40_MACD_HIST_PCT > 0.40$
- $D07_EMA_CROSS < -$
- $D6_BB < -$
- $D30_ATR_PCT > 0.27$
- $D09_VOL_NORMAL > 0.10$
- $D24_BB < -$
- $D2_ATR_PCT > 0.27$
- $0.55 < D06_DOWN < -$

LIME — META (Day 1, most recent window)

Features contributing to META Day 1 return (most negative to most positive):

- $D03_MOMENTUM_20 > 0.54$
- $D4_MACD_HIST_PCT > 0.69$
- $D02_MOMENTUM_20 < -$
- $D04_SPY_RETURN > 0.58$
- $D01_MOMENTUM_20 < -$
- $0.63 < D06_SPY_RETURN > 0.69$
- $D40_VOL_RATIO < -$
- $D13_BB_WIDTH > 0.14$
- $D05_ATR_PCT > 0.22$
- $0.49 < D02_RETURN_30 < -$
- $D21_VOL_RATIO < -$
- $0.43 < D10_MOMENTUM_20 < -$
- $D06_VOL_NORMAL > 0.10$
- $D33_DOWN > 0.79$
- $0.09 < D09_VOL_RATIO < -$

LIME — NVDA (Day 1, most recent window)

Features contributing to NVDA Day 1 return (most negative to most positive):

- $0.22 < D14_ATR_PCT < -$
- $D40_VOL_RATIO > 0.20$
- $0.55 < D05_VOL_RATIO > 0.21$
- $D40_VOL_NORMAL > 0.11$
- $0.12 < D40_BB_WIDTH < -$
- $D42_BB_PCT > 0.22$
- $0.50 < D10_DOWN < -$
- $D29_VOL_NORMAL > 0.10$
- $D40_EMA_CROSS < -$
- $D40_VOL_RATIO < -$
- $D17_RETURN_10 < -$
- $D10_EMA_CROSS < -$
- $D12_VOL_NORMAL > 0.09$
- $D05_MACD_HIST_PCT > 0.74$
- $D4_MACD_HIST_PCT < -$

LIME — TSLA (Day 1, most recent window)

Features contributing to TSLA Day 1 return (most negative to most positive):

- $D08_BB_POS > 0.64$
- $D08_BB_POS > 0.69$
- $D08_BB > 0.62$
- $D08_BB > 0.62$
- $D09_BB > 0.62$
- $0.50 < D40_BB_POS < -$
- $D4_ATR_PCT > 0.17$
- $0.25 < D16_DOWN < -$
- $D49_SPY_RETURN < -$
- $0.30 < D06_MOMENTUM_20 < -$
- $0.43 < D45_BB < -$
- $D11_RETURN_10 < -$
- $0.37 < D08_MOMENTUM_20 < -$
- $D09_MACD_HIST_PCT > 0.77$
- $D11_VOL_RATIO > 0.24$

LIME — NFLX (Day 1, most recent window)

Features contributing to NFLX Day 1 return (most negative to most positive):

- $D21_VOL_RATIO > 0.21$
- $0.48 < D02_BB_POS < -$
- $D05_BB_WIDTH > 0.09$
- $0.44 < D40_EMA_CROSS < -$
- $D11_RETURN_10 < -$
- $D45_VOL_RATIO < -$
- $D49_BB > 0.72$
- $D02_VOL_NORMAL > 0.11$
- $0.49 < D02_MACD_HIST_PCT > 0.69$
- $0.25 < D06_DOWN < -$
- $D13_BB_POS < -$
- $D05_EMA_CROSS > 0.87$
- $D11_RETURN_10 < -$

LIME — INTC (Day 1, most recent window)

Features contributing to INTC Day 1 return (most negative to most positive):

- $D09_RETURN_30 < -$
- $0.49 < D11_BB_POS < -$
- $D11_VOL_NORMAL > 0.09$
- $D09_DOWN < -$
- $0.40 < D06_EMA_CROSS < -$
- $0.40 < D11_RETURN_30 < -$
- $D49_VOL_RATIO < -$
- $D06_BB_POS < -$
- $0.43 < D40_RETURN_10 < -$
- $0.44 < D13_MACD_HIST_PCT < -$
- $0.49 < D02_BB_POS < -$
- $0.31 < D16_BB < -$
- $0.55 < D37_SPY_RETURN < -$
- $D05_RETURN_30 > 0.59$
- $D10_ATR_PCT > 0.50$

Key findings from LIME (Figure 5):

- **AAPL, MSFT:** DOW (day-of-week) is the top local negative driver for both stocks — large-cap tech has strong weekly seasonality patterns the model has learned and relies on locally.
- **GOOGL:** D1 and D2 features (the oldest days in the 60-day window, ~2 months ago) dominate. The model uses longer-range context for GOOGL than any other stock, suggesting stronger long-range momentum dependencies that are unique among the 10 stocks.
- **META:** D53_MOMENTUM_20 is a large negative driver — medium-term momentum from ~7 weeks ago pulls predictions down. Past momentum cycles matter more than recent signals for META.
- **TSLA, AMD:** Recent BB_POS and RSI (D59, D60) dominate and are positive — perfectly consistent with the SHAP beeswarm finding. The model asks: 'where is the price right now within its Bollinger Band?' Both stocks are highly momentum-driven at short range.
- **NVDA, AMZN:** VOL_RATIO drives predictions locally — volume spikes are the strongest signal, reflecting institutional buying/selling pressure in these widely-held stocks.
- **NFLX:** VOL_RATIO from earlier days (D10) dominates — NFLX price moves are preceded by volume signals several days earlier, a lead-lag relationship the model has captured.
- **INTC:** RETURN_5D is a large negative driver — recent 5-day losses strongly predict continued downward movement, reflecting INTC's persistent underperformance trend.

LIME and SHAP results are consistent where expected: TSLA and AMD show BB_POS + RSI as dominant in both analyses. GOOGL's reliance on older historical days is a new finding that only LIME reveals — SHAP averages this out across all time steps. For MSFT/AMZN/NVDA, SHAP identifies VIX_NORM as the global driver while LIME shows volume/BB features locally — both are valid at their respective scopes.

6.3 SHAP vs. LIME Comparison

Property	SHAP (GradientExplainer)	LIME (lime_tabular)
Scope	Global (all predictions)	Local (single prediction)
Theoretical basis	Cooperative game theory	Local linear approximation
Implementation	Exact gradients via TF/Keras	Input perturbation + linear fit
Top feature	VIX_NORM (global market fear)	D60_RSI (most recent RSI)
Stability	Consistent across runs	Slightly variable between runs
Speed	Fast with GradientExplainer	Slow (780 features × 5000 samples)
Best for	Understanding model overall	Explaining one prediction to a client

7. NLP Sentiment Analysis — FinBERT

FinBERT (ProsusAI/finbert) is a BERT model pre-trained on financial news and fine-tuned for Positive/Neutral/Negative sentiment classification. Sentiment score = $P(\text{positive}) - P(\text{negative})$ across up to 20 recent headlines per stock. Scores above +0.15 are POSITIVE, below -0.15 are NEGATIVE, and between ± 0.15 are NEUTRAL.

Ticker	Sentiment	Score	Boost Applied?
AAPL	NEUTRAL	-0.007	No
MSFT	NEUTRAL	-0.077	No
GOOGL	NEUTRAL	-0.016	No
AMZN	NEUTRAL	-0.036	No
META	NEUTRAL	-0.120	No
NVDA	NEUTRAL	+0.147	No
TSLA	POSITIVE	+0.198	+5% boost (aligned with model ▲ UP)
AMD	NEUTRAL	+0.073	No
NFLX	NEUTRAL	+0.023	No
INTC	NEUTRAL	-0.024	No

TSLA is the only stock where sentiment clearly aligns with the model's directional call (both POSITIVE sentiment and model predicts ▲ UP). The platform applies a +5% confidence boost in this case, raising TSLA's confidence from 55% to 60%.

8. Generative AI — Groq + LLaMA 3.1

The Groq API (LLaMA 3.1 8B Instant model) converts numerical model outputs into professional client-facing explanations. Each explanation covers: the predicted price and direction, the model's historical reliability on that stock, the current sentiment signal, and a risk caveat. A Q&A chatbot allows users to ask natural language questions about the predictions.

- **Groq free tier:** 14,400 requests/day with no credit card required
- **Gemini limitation:** Gemini free tier was exhausted immediately due to daily token limits
- **Model quality:** LLaMA 3.1 8B produces high-quality financial summaries suitable for client presentation

9. Live 7-Day Forecast (June 2026)

Ticker	Last Price	Day 7 Predicted	7-Day Return	Direction	Confidence
AAPL	\$200.56	\$201.56	+0.50%	▲ UP	52%
MSFT	\$451.24	\$454.26	+0.67%	▲ UP	53%
GOOGL	\$170.18	\$171.20	+0.60%	▲ UP	56%
AMZN	\$203.10	\$203.81	+0.35%	▲ UP	51%
META	\$634.61	\$640.35	+0.90%	▲ UP	54%
NVDA	\$132.65	\$135.15	+1.88%	▲ UP	52%
TSLA	\$341.04	\$349.27	+2.41%	▲ UP	60%*
AMD	\$110.71	\$112.13	+1.28%	▼ DOWN	53%
NFLX	\$118.80	\$119.50	+0.59%	▲ UP	55%
INTC	\$20.55	\$20.41	-0.67%	▼ DOWN	54%

* TSLA confidence boosted +5% due to aligned POSITIVE FinBERT sentiment.

10. Limitations

- **Modest directional accuracy:** Directional accuracy of 50.8–56.2% is realistic and consistent with academic literature, but modest in absolute terms. Markets are largely efficient and near-unpredictable from price history alone.
- **Classifier UP-bias:** The classifier rarely predicts DOWN for most stocks (Recall = 100% = always predicts UP for 7/10 stocks). Training on 2018–2025 data — predominantly a bull market — produces an upward bias.
- **INTC classifier failure:** INTC's classifier achieved 0% F1 score — it never predicted DOWN. This is a known failure case and must be disclosed when presenting INTC predictions.
- **No fundamental data:** Earnings, P/E ratio, revenue growth, and fundamental data are not included.
- **Sparse sentiment history:** Finnhub free tier provides only recent news; sentiment cannot be backtested on historical data.
- **No macroeconomic features:** Interest rates, inflation, GDP, and Federal Reserve policy signals are absent.

11. Future Improvements

- Transformer architecture with self-attention for better long-range dependency modelling
- Ensemble model (GRU + XGBoost + Transformer) for more robust predictions
- Fundamental data integration: earnings calendar, P/E ratio, revenue growth
- Class-weighted training or threshold tuning to address UP-bias in the classifier
- Attention mechanism to identify which of the 60 past days matter most
- Macro features: Federal Reserve interest rate decisions, CPI, unemployment rate
- Longer training history (10+ years) to include 2008 financial crisis and more market cycles

12. Conclusion

This project successfully builds a complete stock prediction platform covering 10 technology stocks. The dual-model architecture (BiGRU regressor for price + GRU classifier for direction) resolves the model collapse problem that plagued earlier single-model approaches. Price prediction is strong with R^2 of 0.93–0.99 and MAPE of 1–3%. Directional accuracy of 52–56% across 7 of 10 stocks is consistent with state-of-the-art technical-indicator forecasting results.

The XAI layer (Figures 4 and 5) reveals that VIX_NORM and recent RSI are the dominant prediction drivers — providing interpretable, client-friendly explanations for each call. FinBERT sentiment correctly identifies TSLA's positive market sentiment aligning with its bullish model prediction. The Groq-powered LLaMA 3.1 chatbot makes the platform accessible to non-technical users through natural language interaction.

Known limitations — particularly the classifier's UP-bias and INTC's classifier failure — are transparently documented. Future work incorporating fundamental data, macroeconomic features, and Transformer architectures could meaningfully extend predictive accuracy.

13. Technology Stack

Component	Library / Tool
Price Data	Tiingo REST API
Market Context	yfinance (SPY, VIX)
News Headlines	Finnhub API
Deep Learning	TensorFlow 2.20 / Keras
Feature Scaling	scikit-learn MinMaxScaler
XAI — Global	SHAP (GradientExplainer)
XAI — Local	LIME (lime_tabular)
NLP Sentiment	HuggingFace Transformers (FinBERT — ProsusAI)
Generative AI	Groq API (LLaMA 3.1 8B Instant)
Visualisation	Matplotlib
Data Processing	NumPy, Pandas
Model Persistence	joblib, Keras .keras format
Environment Keys	python-dotenv

Disclaimer

All predictions are generated by a machine learning model trained on historical data. They do not constitute financial advice. Past model performance does not guarantee future accuracy. Always consult a qualified financial advisor before making investment decisions.